QuantPulse AI: An Intelligent Stock Market Analysis and Paper Trading Platform Using Artificial Intelligence and Machine Learning

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***Abstract*— Retail traders typically need to assemble price chart movements, indicator values, news headlines and derivatives data from various sources in order to make any one trading decision. This paper introduces QuantPulse AI, a web-based platform that unifies these inputs into one workflow. It calculates technical indicators (RSI, MACD, EMA, SMA, Bollinger Bands, ATR and the Stochastic Oscillator), uses historical prices to train Random Forest and Gradient Boosting models for a seven-day forecast which it displays with a confidence band, scores financial news sentiment and reads Futures and Options (F&O) indicators like Put-Call Ratio, Open Interest and Max Pain. The four streams are combined into a BUY/ HOLD/ SELL signal along with an entry price, a dynamic stop loss and a target price and risk/reward ratio by a weighted Decision Engine. A virtual paper trading terminal has a starting amount of ₹ 10,00,000 and provides an environment where users can test the signals without risk. Written in Python with FastAPI and scikit-learn in the back end and HTML, CSS and JavaScript in the front end. On a live run for RELIANCE.NS the two ensemble models achieved 70% and 72% accuracy; the engine gave a BUY signal with 75% confidence. These numbers are not big and we talk about them openly as what they do and do not tell us.**

Key Words—Machine Learning, Stock market prediction, Random Forest, Gradient Boosting, F&O analysis, paper trading, technical indicators, sentiment analysis and decision support.

I. INTRODUCTION

People who have actually had to make a decision regarding whether or not to purchase a specific stock are aware of the way things are disorganized. The price chart is in one tab, data from the indicator in another tab, a news page for the morning headlines and the options data on a page most beginners never go to, an exchange page. Every source has something useful to say, but seldom they agree with one another, and the one who has to make the choice has to rely on his instincts to make it. It takes years for experienced traders to learn. Newcomers generally don't have those years and training on a live account can be expensive.

This problem has been tackled by machine learning for long and recent efforts have been towards deep sequence models and transformers with text sentiment [1]–[3]. The studies generally report forecasting error or directional accuracy. They rarely ask the user what they should do with the output, how to get that forecast to reconcile with another that shows the opposite, or where to experiment with the output for a safe decision.

That's the space where QuantPulse AI has been designed. The platform does not halt when the price is forecasted; instead, the forecast is one of four variables. They each provide a vote of sorts with a weighting, a seven day machine learning trajectory, news sentiment and F&O positioning, and then a Decision Engine takes the combined vote and translates it into a signal with definite trade levels. The paper trading terminal then completes the circle; after a suggestion, it can be tried with virtual money, and then evaluated at the end.

The contributions of this work are: (1) an integrated analysis pipeline that includes technical, ML, sentiment and F&O evidence, (2) a dual-ensemble forecaster with an increasing confidence band over a multi-day time series, (3) a weighted Decision Engine that returns a BUY/HOLD/SELL with entry, stop-loss, target and risk/reward values, (4) a risk-free paper trading and tracking module, and (5) a single web dashboard that brings all this together for a non-expert.

II. PROBLEM STATEMENT

Price, volume, news and derivatives data are created in large quantities on stock markets each trading day and reading such data by hand is slow. The investor must make a judgment about the price action, indicator readings, news tone and the positioning of Futures and Options. Most of the tools that are available will be showing these alone so that the user will have to mentally add them together, and the only tools that do predict a price are going to be giving the price without any recommendation or level of risk. Beginners are the worst hit, as there is no safe place to practise and they have no easy way of knowing which signal to follow when there are conflicting signals. There needs to be one platform which is explainable, multi-faceted in analyzing the market, gets that analysis

translated into an actionable recommendation, and provides an environment in which you can test it out virtually.

III. OBJECTIVES

Development of an AI based analysis system for real-time and historical data.

Analyse trend using technical indicators and machine learning models.

Include F&O sentiment analysis and F&O analysis (PCR, Open Interest, Max Pain).

Predict the future price movements of the stock using a historical data.

Put all of the above into a single Decision Engine and get the results of BUY, HOLD or SELL.

Simulate a virtual environment for paper trading to practice without risk.

IV. RELATED WORK

Moodi et al. combined technical indicators and sentiment within a CNN-LSTM model and achieved higher accuracy in the prediction of movements compared to the prediction from price information alone [1]. Their finding reinforces our philosophy which is to incorporate indicators and sentiment into the same decision, but they do not include an F&O layer, a recommendation layer or a trading environment. Transformer-based work has been more successful in terms of model capacity: a Node Transformer with BERT sentiment [2] and a combination of LSTM, transformer attention, and federated learning with sentiment [3] both achieve high accuracy, albeit at the cost of complex models and high data and compute requirements. A comparison of LSTM, GRU, CNN and Prophet in the task of predicting news-driven forecasting reveals that there is indeed actual signal in the headlines [4], and a wider survey of classical machine-learning and deep-learning pipelines on historical prices confirms that most pipelines focus on forecasting instead of other tasks [5].

What is striking in these papers are two points. First, the model is getting more complex and less usable: few of them make it clear how the forecast becomes an action. Second, despite the fact that Put-Call Ratio (PCCR) and Open Interest (OI) are two widely used indicators, there is a dearth of derivatives data. It is no wonder that QuantPulse AI takes the opposite approach. It employs two well-known ensemble learners [6, 7] that can be executed on an average laptop, and allocates the remaining effort to integration, the decision layer and paper trading. The contrast is summarised in Table I.

TABLE I
COMPARISON WITH EXISTING APPROACHES

Work	Approach	Gap
[1] 2024	Indicators + sentiment, CNN-LSTM	No F&O, decision engine or trading
[2] 2026	Node Transformer + BERT	Complex; large data needs
[3] 2026	LSTM + attention + federated	Compute-intensive
[4] 2024	LSTM/GRU/CNN/Prophet + news	No F&O or paper trading
[5] 2024	ML/DL on price history	No sentiment, F&O or trading
This work	RF + GB, indicators, news, F&O	Integrated engine + paper trading

V. PROPOSED SYSTEM

The proposed architecture consists of five high level layers as depicted in Fig.1. The data-source layer fetches the real-time quotations and OHLCs, historical prices and volumes, financial-news headlines, and F&O data like PCR and Open Interest. Information is collected and preprocessed in a collection and preprocessing layer, where it is cleaned and validated, aligned and normalised. Moving average, RSI, MACD, ATR, Bollinger bands are then created using feature engineering.

There are four analysis branches on the processed data, namely technical analysis, machine-learning forecasting, news sentiment analysis, and F&O analysis. The results of these are fed into the Decision Engine. This generates the output that is then passed to the Future Stock Prediction module and Paper Trading module which then present the output on the web dashboard.

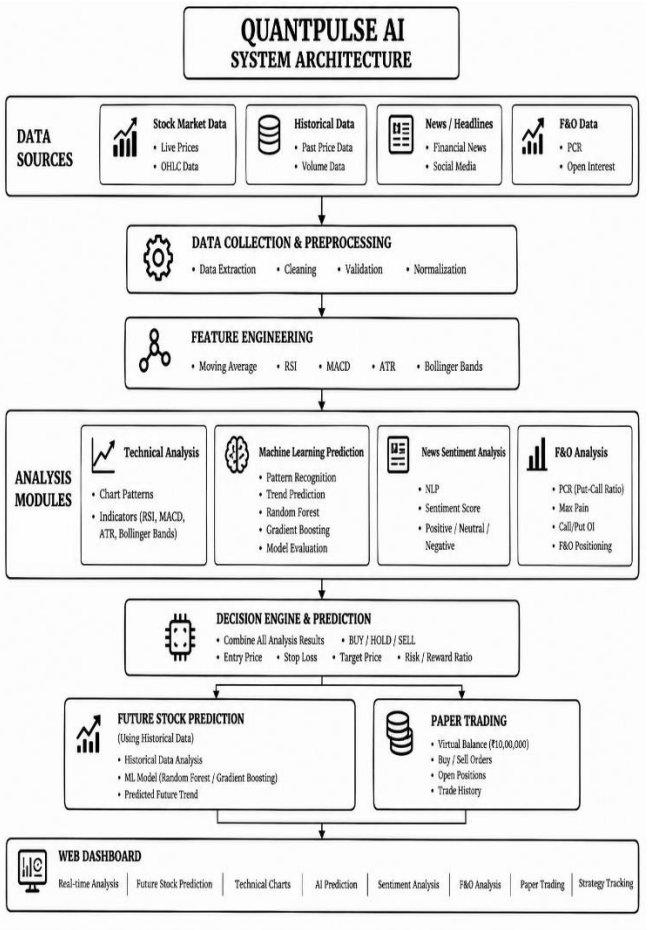


Fig. 1. System architecture of QuantPulse AI.

Branches are purposely cut apart. If one part of the external feed fails, the other parts of the analysis continue to work and the Decision Engine can still leverage the information that is available.

VI. SYSTEM MODULES

Table II lists the ten modules of the application.

TABLE II
APPLICATION MODULES

Module	Function
Data Collection	Retrieves market, historical, news, and F&O information.
Preprocessing	Cleans incoming data and prepares model-ready features.
Technical Analysis	Calculates indicators for trend, momentum, and volatility assessment.
AI/ML Prediction	Learns patterns from historical features and estimates direction.
Historical Prediction	Constructs a seven-day projected price path.
Sentiment Analysis	Classifies recent financial headlines as positive, negative, or neutral.
F&O Analysis	Examines PCR, Open Interest, and Max Pain.
Decision Engine	Combines the analysis streams and produces BUY, HOLD, or SELL.
Paper Trading	Simulates orders and portfolio changes with virtual capital.
Web Dashboard	Presents analysis, forecasts, signals, and trading results.

VII. METHODOLOGY

A. Data Collection

On a particular symbol like RELIANCE.The platform receives fresh OHLCV data, a longer time to backfill the model, new financial news, and the information associated with the relevant option-chain. To generate features from external feeds, it is necessary to preprocess these feeds, as they may not be structured in the same way, which involves removing incomplete data, normalising data, checking data ranges, and aligning timestamps. This is an important integration step when applying it in practice, because small differences between feeds can lead to calculations downstream failures, or clashes.

B. Technical Indicators

The indicator set includes trend, momentum and volatility. RSI is calculated over 14 periods and is used as a momentum measure: RSI is calculated over 14 periods and is used as a momentum measure, where RS is the ratio of average gain to average loss:

$$RSI = 100 - 100 / (1 + RS),\ RS = \text{avg gain} / \text{avg loss} \quad (1)$$

The EMA values are given more weight than those values from the previous observations, where α is equal to $2/(N + 1)$:

$$EMA_t = \alpha \cdot P_t + (1 - \alpha) \cdot EMA_{t-1},\ \alpha = 2/(N + 1) \quad (2)$$

Macd is calculated by using the 12-EMA and 26-EMA, and is evaluated using a 9-EMA. The Bollinger Bands of 20 periods and 2 standard deviations are used. ATR is a measure of price volatility, and SMA and the Stochastic Oscillator offer extra trend and momentum details. The supported interpretation is a negative MACD with price closing below the corresponding EMAs is bearish, and RSI readings below 30 or above 70 are stretched conditions.

C. Machine-Learning Forecast

Two types of supervised ensemble learners: Random Forest and Gradient Boosting [6, 7] are used in the forecasting stage. Random Forest, on the other hand, combines predictions from many decision trees and Gradient Boosting sequentially fits trees to errors of previous trees. Both are done using scikit-learn [8].

A simple cross-check is achieved by using two models with different learning behaviour. Their outputs are fed into an ensemble and historical drift and volatility estimates extended the forecast for seven days. The further away the forecast, the wider the range of uncertainty, which means there is less certainty further into the future. The system also sends predictions of the direction, direction confidence and projected next close.

D. News Sentiment

The selected symbol is used to retrieve recent headlines that are processed by a natural-language classification step. Every headline is classified as “positive,” “negative” or “neutral.” The labels are summed to get a sentiment score ranging from −1 to +1. The user can see the aggregate score and individual class counts on the dashboard, as well as the headline list, to view the evidence used for the score.

E. F&O Analysis

Put-Call Ratio: It is defined as the ratio of Put Open Interest to Call Open Interest:

$$PCR = \Sigma \text{ Put OI} / \Sigma \text{ Call OI} \quad (3)$$

Any value near 1 is seen as a fairly neutral price, and values that are materially above or below are considered as evidence from a resistance or support area. Max Pain is the strike price having the least loss in total value of the option-writers at expiry, and sometimes can be viewed as a level close to expiry. The module also presents resistance and support for the call side, as well as support and resistance for the put side and the Open Interest for strike-wise.

F. Decision Engine

The four analysis streams are transformed into signed contributions, and then these contributions are weighted and aggregated according to pre-determined weights. If the composite score is positive enough, it will be a BUY; if it is negative enough, it will be a SELL; and if it is in the middle, it will be a HOLD. A confidence percentage is assigned to the same score. If an actionable signal is generated, the engine will also determine an entry reference, a volatility stop-loss, a target, and the risk/reward ratio.

G. Paper Trading & Strategy Tracking

The paper trading feature mimics the market orders with a virtual balance of ₹ 10,00,000. The simulated portfolio is

stored in SQLite, and the positions, cash, and trade history are stored as well, so that the simulated portfolio will be available after restarting the application. The strategy-tracking feature provides metrics like win rate, the number of trades, profit factor, and realised profit/loss, enabling users to see what kind of performance the generated signals had throughout time.

VIII. SYSTEM REQUIREMENTS

Hardware requirements: A computer with a minimum of 4 GB of RAM.Software requirements: Windows 10/11; Python; FastAPI; scikit-learn; Pandas; NumPy; Matplotlib and HTML/CSS/JavaScript charts; SQLite; Visual Studio Code. The recommended hardware is an Intel Core i5 processor or equivalent, with a minimum of 8GB RAM, around 256GB SSD drive space and a strong internet connection.

TABLE III
SOFTWARE AND HARDWARE REQUIREMENTS

Item	Specification
Operating system	Windows 10/11
Language	Python
Backend	FastAPI
Machine learning	scikit-learn
Data processing	Pandas, NumPy
Visualisation	Matplotlib; HTML/CSS/JS charts
Database	SQLite
IDE	Visual Studio Code
Processor / RAM	Intel Core i5 or equivalent / 8 GB min.
Storage / Internet	256 GB SSD / stable connection

IX. IMPLEMENTATION

The backend is developed using FastAPI as a service, and has endpoint groups for each analysis module. The server also retrieves the data and then calculates the Technical, Machine Learning, Sentiment and F&O calculations and returns the results as consolidated JSON to the Decision Engine, who then returns a consolidated JSON response. Data processing is done with the help of the packages Pandas and NumPy; the ensemble models are part of the scikit-learn package of NumPy; the paper trading ledger is handled by SQLite.

The front end is based on HTML, CSS and JavaScript, and is themed in dark colors. The application runs locally in the prototype at 127.0.0.1:8000. All the information is displayed on the dashboard, which is divided into eight sections: real-time analysis, stock prediction for the future, technical charts, AI prediction, sentiment analysis, F&O analysis, paper trading, and strategy tracking. The quick select symbols include NVDA, AAPL, TSLA, MSFT, RELIANCE and TCS. The steps are essentially the following: Determine the stock, review the composite recommendation and supporting data, view the forecast, and (optionally) order the stock virtually.

TABLE VII

Each tab has a specific purpose.There is a purpose behind each tab on the DASHBOARD.

Tab	Purpose
Real-time Analysis	Composite signal, trade levels, weightings
Future Stock Prediction	7-day trajectory and day-wise table
Technical Charts	Candlesticks, EMA 20, Bollinger Bands
AI Prediction	Direction, confidence, projected close
Sentiment Analysis	Scored and labelled news headlines
F&O Analysis	PCR, Max Pain, strike-wise Open Interest
Paper Trading	Virtual order ticket and portfolio
Strategy Tracking	Win rate, profit factor, realised P&L

X. RESULTS AND DISCUSSION

The entire pipeline was assessed with the RELIANCE.NS on 7 October 2026. In the recorded run, the market price of the share is Rs. 1209.00, which is down by ₹9.00 from the previous day (-0.74%). The daily range of the day was ₹1203.50 to ₹1220.80 and the volume of the day was about 7.50M. The following observations are related to that one live execution.

A. Seven-Day Price Prediction

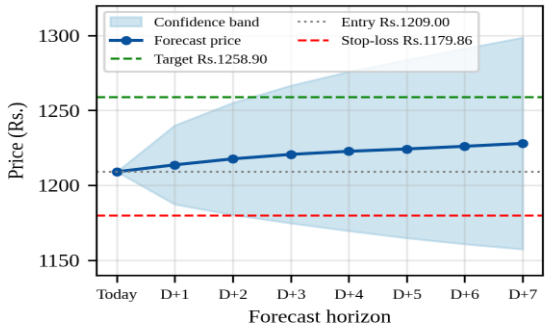


Fig. 2. Predicted 7-day price trajectory with confidence band and trade levels.

As observed from Fig. 2 and Table IV, the mean average of the generated trajectory is observed to gradually rise from ₹1209.00 to about ₹1227.95 at the 7th forecast day, which corresponds to a change of approximately 1.6%. The seven forecast points are all marked in bull market. The uncertainty interval is about ±₹26 on the first day, and increases to about ±₹70 by the seventh day. The widening is significant because it conveys the message that longer-term values are more uncertain. The projected range is broad with the Decision Engine at the high end of the range at ₹1258.90, which is not reflective of the central forecast.

The reported daily forecasts are: Day +1, ₹1213.63 with a range of ₹1187.26–₹1240.00; +2, ₹1217.63 with ₹1180.22–₹1255.04; +3, ₹1220.61 with ₹1174.68–₹1266.54; +4, ₹1222.66 with ₹1169.53–₹1275.79; +5, ₹1224.23 with

₹1164.75–₹1283.71; +6, ₹1226.00 with ₹1160.75–₹1291.25; and +7, ₹1227.95 with ₹1157.31–₹1298.59.

TABLE IV
DAY-BY-DAY FORECAST FOR RELIANCE.NS (₹)

Day	Forecast	Confidence range
+1 (08-Oct)	1213.63	1187.26 – 1240.00
+2 (09-Oct)	1217.63	1180.22 – 1255.04
+3 (10-Oct)	1220.61	1174.68 – 1266.54
+4 (11-Oct)	1222.66	1169.53 – 1275.79
+5 (12-Oct)	1224.23	1164.75 – 1283.71
+6 (13-Oct)	1226.00	1160.75 – 1291.25
+7 (14-Oct)	1227.95	1157.31 – 1298.59

B. Decision Engine Output

The Decision Engine made a BUY decision with a confidence level of 58.6% and a composite score of +17.9. The suggested entry price was Rs.1209.00, the dynamic stop loss level was Rs.1179.86, the target price was Rs.1258.90 and the potential risk & reward ratio was 1.67:1.

The contribution was weighted as shown in Fig. 3, which also helps to understand the final outcome of not being uniformly bullish. Technical analysis was the main negative contributor with a score of −15, while machine learning and news sentiment were the primary positive contributors with scores of +64.9 and +24.6, respectively. The recommendation was thus predominantly based on the machine-learning forecast with positive news and with opposing evidence from the technical indicators.

The supporting readings listed in Table V are: RSI 41.8, MACD -24.39/-25.05, price below EMA20 at ₹1221.40, price below EMA50 at ₹1256.39, machine-learning direction UP with +1.98% and 82% direction confidence, projected close ₹1232.33, news sentiment +0.25 (3 positive, 1 negative, 2 neutral) and F&O PCR 1.03 with Max Pain at ₹1200.

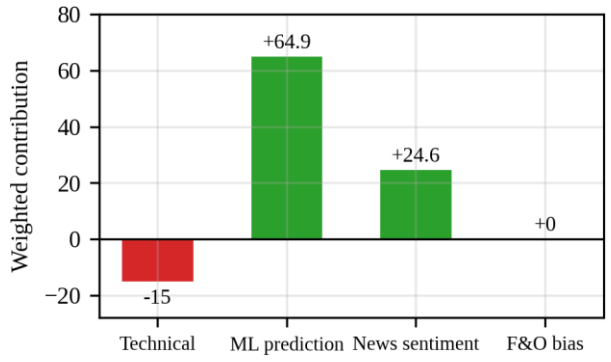


Fig. 3. Weighted contribution of each analysis stream to the composite signal.

TABLE V
EVIDENCE BEHIND EACH CONTRIBUTION

Stream	Key readings	Score
Technical	RSI 41.8; MACD −24.39 / −25.05; price below EMA20 (₹1221.40) and EMA50 (₹1256.39)	−15
ML	Direction UP (+1.98%); confidence 82%; projected close ₹1232.33	+64.9
News	Bullish (+0.25); 3 positive, 1 negative, 2 neutral	+24.6
F&O	PCR 1.03 (neutral); Max Pain ₹1200	0

C. Model Accuracy

Random Forest and Gradient Boosting achieved a Directional accuracy of 70% and 72% respectively. The ensemble accuracy of their two ensembles was 75% (Fig. 4 and Table VI). They are just slightly higher than a 50% directional baseline, and should be interpreted with some caution as evidence of reliable prediction on their own. There is increased noise in market direction and a limited current evaluation. Unlike traditional trading oracles, the forecast in QuantPulse AI is not an autonomous entity but instead serves as one of the multiple sources of evidence.

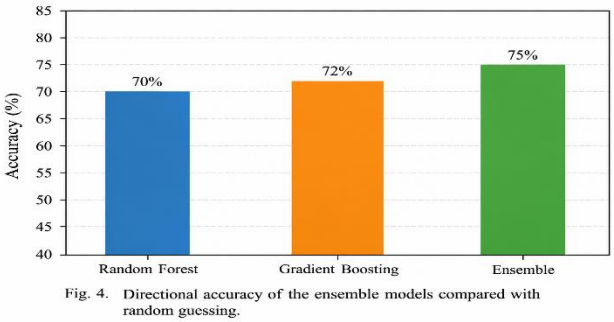


Fig. 4. Directional accuracy of the ensemble models compared with random guessing.

Fig. 4. Directional accuracy of the ensemble models compared with random guessing.

TABLE VI
MODEL PERFORMANCE (RELIANCE.NS)

Model	Accuracy
Random Forest	70%
Gradient Boosting	72%
Ensemble	75%

D. Technical, Sentiment and F&O Readings

The technical indicators were responsible for the negative contribution. During the run, RELIANCE.NS had been on the down side, trading below both the 20 period EMA and 50 period EMA around ₹1300 now down to ₹1150. RSI was at 41.8 and MACD was in negative. Bollinger Bands stretched between ₹1158.8 and ₹1284.3 and ATR (19.96) suggested high volatility. The overall picture indicates that the BUY signal could be a reversionary move and not the confirmation of a bull market trend.

The news part rated the sentiment as 'bullish' (+0.25). The headlines analysed mentioned a Q3 revenue beat, which was linked to cloud and AI growth, a standard proxy filing, and institutional build-up. The neutral F&O data resulted in a PCR of "1.03". The Open Interest in calls and puts for the expiry on 8 October 2026 stood at around 928k and 957k respectively, with the strike price of max pain at ₹1200 (Fig. 5). The spot value was near to Max Pain and the F&O branch *did not make a directional contribution.*

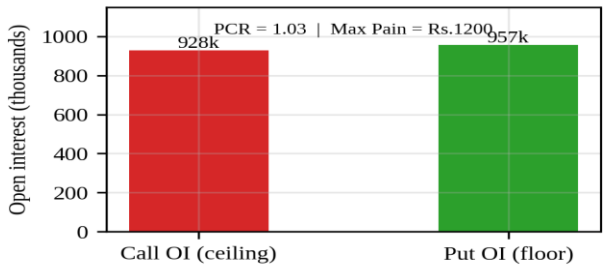


Fig. 5. Call versus put Open Interest for the nearest expiry.

E. Paper Trading and Strategy Tracking

On the paper-trading terminal, the user can choose to BUY or SELL, can enter the quantity and can preview the order to be executed before confirming it. Quantity shortcuts include +10, +50, and +100. So, for instance, if you enter a simulated order of 10 shares at ₹1209.00, then the notional value is ₹12,090.For instance, a 10-share simulated order at ₹1209.00 will have a notional value of ₹12,090. The portfolio view shows cash, net worth, unrealised P&L, open positions, and trade history, while the strategy tracking reports shows win rate, trade count, profit factor, and realised P&L. These numbers apply to a demonstration account, and thus do not indicate the statistical quality of the strategies.The numbers apply to a demo account and therefore are not measures of statistical strategy quality.

XI. TESTING AND VALIDATION APPROACH

Testing is currently mostly a functional testing exercise. The values returned by the analysis modules were checked, dashboard displays were compared to backend responses, increasing uncertainty band over time, and paper trading actions were continually updating cash, positions, and history. These tests do not prove profitable but rather show end-to-end working.

A better predictive assessment should use chronological training and testing, train models using data from previous time points and test the models using data from future time points not used for training. Multiple walk-forward tests done on several stocks would minimize the risk of drawing conclusions based on one example. Precision, recall, F1-Score, and confusion matrix (for the direction classes) should be included in the evaluation. Trading signals should also be backtested with account for transaction costs and should also provide returns, drawdown, and Sharpe ratio. Decision weights should be chosen one period and measured on another. The accuracy reported should be viewed with a degree of scepticism until this process is complete.

XII. CHALLENGES AND LIMITATIONS

The challenges faced during development included addressing different data formats, technical, ML and sentiment consistency, choice of weights for the Decision Engine, integration with the dashboard, and consistency of the simulated trading behaviour.

There are important limitations to the current implementation. The accuracy of the forecast is fairly low and has only been shown to work on one stock in the Evaluation reported. The Decision Engine's weights are set by hand and not based on past results. Sentiment quality is based on the headlines provided by the source. The current signal only takes into account a single horizon and excludes transaction costs and slippage. Therefore, the platform should not be considered as a piece of investment advice, but as a learning and decision support.

Considerable care should be taken in presenting a Responsible indication, as a confident BUY or SELL indication can be misinterpreted as a guarantee. The interface then displays the confidence value, the individual stream contributions and the stop-loss information, and it encourages the people to practise using paper trading. Simulated execution can also be more favorable than actual execution in that it doesn't faithfully simulate the liquidity constraints, brokerage costs, slippage, or psychological impact of financial losses.

XIII. CONCLUSION AND FUTURE WORK

The combination of tested technical indicators, ensemble machine-learning forecasts, financial-news sentiment and F&O information is integrated via a weighted Decision Engine, as is demonstrated in the case of QuantPulse AI. The platform then links the created suggestion to a paper trading space, where customers can try out signals without risking any genuine money.

The RELIANCE.NS demonstration demonstrated the interaction of these elements: the platform issued a BUY recommendation with a 58.6% confidence, a slowly increasing 7-day forecast with a widening uncertainty range and trade levels that matched a 1.67:1 risk/reward ratio. Meanwhile, the 75% ensemble accuracy suggests that the recommendation should be taken with a grain of salt as there is disagreement between the bearish technical evidence and bullish machine-learning output.

Further improvements should start with a thorough walk-forward testing of several stocks and a longer time period, such as precision and recall, and transaction-cost-aware back-testing. Other extensions can include LSTM and Transformer time-series models, Explainable AI for signal justification, data-driven instead of manually selected weights, portfolio optimisation, position sizing, improved risk management, real-time updates, other F&O measures, and more comprehensive paper-trading analytics.

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